

Constitutive Relevance, Mutual Manipulability, and Fat-Handedness

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ABSTRACT

The first part of this article argues that if Craver's ([2007a], [2007b]) popular mutual manipulability account (MM) of mechanistic constitution is embedded within Woodward's ([2003]) interventionist theory of causation—for which it is explicitly designed—it either undermines the mechanistic research paradigm by entailing that there do not exist relationships of constitutive relevance, or it gives rise to the unwanted consequence that constitution is a form of causation. The second part shows how Woodward's theory can be adapted in such a way that MM neither undermines the mechanistic paradigm nor reduces constitution to causation. However, it turns out that this modified theoretical embedding of MM makes it impossible to produce empirical evidence for constitutive relations. The article ends by suggesting an additional criterion—the fat-handedness criterion—which, when combined with MM generates indirect empirical evidence for constitutive relevance.

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1 Introduction

To mechanistically explain the behaviour of a system, S , it must be determined which of its parts are constitutively relevant to S 's behaviour, and what causal structure regulates the behaviour of those parts (Craver [2007b], Section 4). While there are various well-tried frameworks for uncovering causal structures (for example, Spirtes *et al.* [2000]; Pearl [2009]), the problem of how to

account for relations of constitutive relevance has, at least until Craver's influential book *Explaining the Brain* ([2007b]), attracted almost no attention. Even though Craver's mutual manipulability approach MM to constitutive relevance has been critically examined repeatedly (see, for example, Harbecke [2010]; Couch [2011]; Leuridan [2012]; Glauer [2012], Section 3.3.2; Schindler [forthcoming]; Franklin-Hall [unpublished]), MM is the dominant account of constitutive relevance—notably because of its considerable intuitive plausibility and alleged closeness to scientific practice.¹

In a nutshell, according to MM, the behaviour of a spatio-temporal part, X , of a system, S , is constitutively relevant to S 's behaviour if and only if the behaviours of X and S are mutually manipulable (Craver [2007a], pp. 15, 17). By 'manipulability', Craver ([2007b], Section 4.8.3) means the possibility of a change in behaviour brought about by an intervention in the manner of (Woodward ([2003])). The aim of this article is to scrutinize to what extent MM adequately identifies relations of constitutive relevance.

In the first part, we demonstrate that when MM is embedded within Woodward's ([2003]) interventionist theory of causation—for which it is explicitly designed—it either undermines the mechanistic research paradigm by entailing that there do not exist cases of constitution, or it gives rise to the consequence that constitution is a form of causation, a consequence that most mechanists reject. These unwanted ramifications of MM are symptoms of a more general problem that Woodward's interventionist theory faces when it is applied to structures containing non-causally dependent variables, such as variables that are related in terms of logical entailment, definition, supervenience, and the like (see, for example, Baumgartner [2010], [2013]).

In a recent article, Woodward ([forthcoming]) recognizes this problem and proposes a modification of his original theory, which he claims produces adequate results even if applied to structures featuring non-causally dependent variables. In the second part of the article, we reconsider the prospects of MM when embedded in this recent version of interventionism. It turns out that the latter does not solve the problems of MM. We then propose a further (temporal) refinement of the interventionist theory and show that, when embedded in this temporal variant of interventionism, MM neither undermines the mechanistic paradigm nor reduces constitution to causation.

Yet, although MM works as intended within this refined interventionist framework, the final part of the article reveals that the mere mutual

¹ Authors that draw on MM include Illari and Williamson ([2011]), Casini *et al.* ([2011]), Soom ([2011]), Kaplan ([2012]) (who is particularly enthusiastic about MM), Gillett ([2013]), Irvine ([2013], Section 6), and Zednik ([forthcoming]). According to Levy ([2009], p.141), MM is one of the main achievements of Craver's book. Even Fagan ([2013], p. 100), who agrees that MM has considerable weaknesses, acknowledges that 'Craver's is the best-developed account of MEX [mechanistic explanation] in biology currently on offer'.

manipulability of the behaviour of a part, X , and of the behaviour of a macro-system, S , does not provide empirical evidence for the former being a constituent of the latter—a finding which contradicts the standard opinion in the literature (see, for example, Craver [2007b], pp. 132, 159; Couch [2011], p. 381). The article ends by developing an additional criterion, the fat-handedness criterion. We show that constitution provides the best available explanation for systems satisfying both mutual manipulability and fat-handedness. Mutual manipulability and fat-handedness together, thus, lend abductive support (for a general approach, see Schurz [2008]) for the existence of constitutional relationships.

2 Mechanisms and Constitutive Relevance

In recent years, mechanisms have begun to play an increasingly important role in the philosophy of science, especially when it comes to explaining the behaviour of macro-systems in the special sciences. Among the most frequently cited mechanistic theories are (Machamer *et al.* [2000]; Glennan [2002]; Bechtel and Abrahamsen [2005]). Craver ([2007a], p. 6) identifies the following consensus among mechanists: ‘mechanisms are entities and activities organized such that they exhibit the phenomenon to be explained. The entities are the parts. The activities are what they do’. Plainly, this characterization raises numerous follow-up questions, for instance, as to the nature of the entities and activities that figure in mechanisms. For our subsequent purposes, however, the details of a comprehensive account of the notion of a mechanism are not relevant. In what follows, we sidestep all complications that come with spelling out that notion. Instead, we focus on a simple exemplary mechanism, α , whose relevant features are so generic and unspecific that none of our results will hinge on the concrete structure of α . α is depicted in Figure 1.

α is a whole consisting of several parts. Mechanisms are systems whose behaviour can be described at two different levels of specification: the level of the system as a whole (the macro-level) and the level of the system’s causally interacting constituents (the micro-level). The label ‘mechanism’ is applied both to the whole and to the causal organization of its parts (Craver [2007b], pp. 6–7). A mechanism’s macro-behaviour is often also called a ‘phenomenon’. To avoid terminological complications, we subsequently simply speak of a mechanism’s macro- and micro-levels. The macro-level of α is represented by the upper and the micro-level by the lower ellipse in Figure 1.

Macro- and micro-levels of mechanisms are typically represented by means of individual constants S , X_1 , X_2 , $X_3, \dots$, where S refers to the system as a whole and X_i refers to a constitutively relevant part of S , and variables Ψ , Φ_1 , Φ_2 , $\Phi_3, \dots$, where the values of Ψ represent the possible behaviours of S and the values of Φ_i the possible behaviours of X_i (Craver [2007b], pp. 153–60).

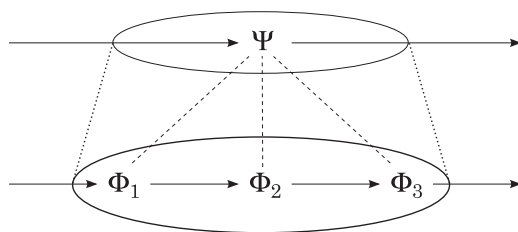


Figure 1. Exemplary mechanism α . Ψ represents the mechanism's behaviour at the macro-level; Φ_1 , Φ_2 , and Φ_3 stand for the behaviours of α 's constitutively relevant parts. The arrows symbolize direct causation (see Section 3), the dotted lines indicate that the macro-level supervenes on the micro-level, and the dashed lines represent constitution. For instance, α can be taken to represent the mechanism constituting the nastic movement of a mimosa (modelled by Ψ), where Φ_1 models the release of potassium ions in the pulvini cells, Φ_2 the cells' turgor pressure, and Φ_3 whether their parenchyma tissue collapses.

A theory of constitutive relevance must spell out under what conditions X_i 's Φ_i -ing, that is, X_i 's realizing of one of the values of Φ_i , is constitutively relevant to S 's Ψ -ing, that is, S 's realizing of one of the values of Ψ .

In what follows, we represent ' S 's Ψ -ing' by means of the specific variable $\Psi(S)$, whose values from the space $val(\Psi(S))$ correspond to the possible behaviours of the system S as a whole; and ' X_i 's Φ_i -ing' we represent by means of the specific variable $\Phi_i(X_i)$, whose values from the space $val(\Phi_i(X_i))$ correspond to the possible behaviours of S 's part X_i .² Contrary to generic variables that are used to represent behaviours as exhibited by any entities, specific variables represent behaviours as exhibited by particular entities (Spohn [2006]). That is, $\Phi_i(X_i)$ taking one of its values—for instance $\Phi_i(X_i) = \phi_1$ —means that entity X_i exhibits behaviour ϕ_1 , and analogously for $\Psi(S) = \psi_1$. For ease of reading, we will mostly abstain from making the specificity of $\Psi(S)$ and $\Phi_i(X_i)$ explicit; that is, we will simply write Ψ for $\Psi(S)$ and Φ_i for $\Phi_i(X_i)$.

Even though concrete interpretations of the variables in our exemplary mechanism α are not critical for our argument, the latter will be more easily accessible relative to a concrete background. Thus, for instance, $\Psi(S)$ could be taken to represent the nastic movement of a mimosa. The spatio-temporal parts of the mimosa (S) that are involved in this phenomenon are the potassium ions in the pulvini cells (X_1), the latter's turgor pressure (X_2), and the cells' parenchyma tissue (X_3). More specifically, the phenomenon $\Psi(S)$ is constituted by a sudden release of potassium ions in the pulvini cells

² In principle, the value spaces $val(\Psi(S))$ and $val(\Phi_i(X_i))$ can be infinite.

$\Phi_1(X_1)$, which causes a drop in the cells' turgor pressure $\Phi_2(X_2)$, which, in turn, leads to a collapse of their parenchyma tissue $\Phi_3(X_3)$.³

Before we can turn to Craver's mutual manipulability theory of constitution, we have to assemble the essential features of the relation of constitutive relevance. It has three uncontroversial features and one feature that is contested. Let us begin with the former. It is generally agreed that the macro-behaviours of a mechanism supervene on their constitutively relevant micro-behaviours, which amounts to the claim that there cannot be a difference in macro-behaviours without a difference in micro-behaviours (Craver [2007b], p. 153), or contrapositively put:

SUP: Every change in the supervening behaviour is necessarily accompanied by a change in the supervenience base.

Note that SUP only renders it impossible to change the macro-level of a mechanism without changing its micro-level. By contrast, the latter may well be changeable without a corresponding change in the former.

While the mechanistic literature is clear with respect to SUP, it is not determinate about what the supervenience base of a mechanism's macro-level is. In consequence, it is somewhat dubious what SUP entails when applied to mechanism α . Craver ([2007a], p. 15), for example, claims that the macro-level of a mechanism 'supervenes on the organized activities of all of the components in the mechanism' (similarly, see Soom [2012], p. 661). Since the 'activities' in mechanisms are typically represented by causal arrows, this quote suggests that α 's macro-behaviour Ψ supervenes on

(a) the variables in $V_1 = \{\Phi_1, \Phi_2, \Phi_3\}$ and the causal structure $\Phi_1 \rightarrow \Phi_2 \rightarrow \Phi_3$.

In that case, SUP entails that any change in Ψ must either be accompanied by a change in the values of at least one variable in V_1 or by a change in the underlying causal structure. Elsewhere, by contrast, Craver ([2007b], p. 153) says that 'supervenience [...] is a relation between a phenomenon and the corporate behavior of the organized components'. As behaviours of components are represented by value configurations of the variables in V_1 , this quote suggests that α 's macro-behaviour Ψ supervenes on

(b) the variables in $V_1 = \{\Phi_1, \Phi_2, \Phi_3\}$.

In that case, it follows from SUP that any change in Ψ must be accompanied by a change in at least one of the variables in V_1 .

Obviously, candidate supervenience bases (a) and (b) are logically dependent—even more so against the background of a difference-making theory of causation as Woodward's ([2003]), which Craver ([2007b],

³ For a more detailed description see, for instance, (Jaffe *et al.* [2002]).

Section 3) explicitly endorses. According to Woodward, the causal structuring among a set of variables is determined by the value configurations of those variables under possible interventions (see Section 3). There cannot be a change in a causal structure without a change in involved variables. Hence, macro-phenomena that supervene both on their constituents' behaviours and activities in the vein of (a), supervene on the behaviour of their constituents alone—as expressed in (b). That is, by following Craver ([2007b]) in adopting Woodward's interventionist theory of causation as our conceptual background, we can safely conclude that the macro-behaviour of mechanism α represented by Ψ , in any case, supervenes on the variables in V_1 . We shall hence settle for candidate supervenience base (b).

The mechanistic literature is also undecided in regard to the ontological question of whether a mechanism's macro-properties reductively or non-reductively supervene on their constituents. Only a minority of authors take a clear stance on this issue. For instance, Eronen ([2011], Section 10, [2012]), Glennan ([1996], pp. 61–2), and Kistler ([2009]) favour a pluralistic ontology that comprises both macro- and micro-properties, while Fazekas and Kertesz ([2011]) and Soom ([2012]) advance the ontological reducibility of the former to the latter. By contrast, most mechanists sidestep metaphysical issues because they take the framework to be metaphysically neutral (see, for example, Craver [2007b], p. 196).

As the main focus of this article is on the question of identifying constitutive relevance, we will leave the ontological relation between our exemplary mechanism's macro- and micro-levels open. Rather, we shall adopt a perspective of causal and mechanistic modelling that views the variables Ψ , Φ_1 , Φ_2 , and Φ_3 as pure modelling devices. Ψ , Φ_1 , Φ_2 , and Φ_3 are different random variables such that changes in the values of Ψ supervene on changes in $V_1 = \{\Phi_1, \Phi_2, \Phi_3\}$.

Apart from SUP, the relation of constitutive relevance is commonly characterized by spatio-temporal congruence (SC) (Craver [2007a], pp. 5–6) and non-redundancy (NR) (Craver [2007b], p. 157):

SC: The constituents of a mechanism are spatio-temporal parts of the mechanism's macro-level.

NR: A spatio-temporal part of a mechanism's macro-level that under no circumstances makes a difference to the mechanism's macro-behaviour is not a constituent of that mechanism.

Both of these features need to be spelled out in more detail. But as neither of them will be at the centre of our discussion, we can confine ourselves to referring to further reading: Leuridan ([2012], Section 6) and Franklin-Hall ([unpublished], Section 4), for example, expound problems that come with SC, and Harbecke ([2010], p. 275) and Couch ([2011], Section 5) provide a specification of NR in terms of minimal sufficiency and necessity.

SUP, SC, and NR are the uncontroversial features of the relation of constitutive relevance. There is one additional feature in regard to which opinions diverge: on the one hand, Craver and Bechtel ([2007]) contend that constitutive relevance is a non-causal form of dependence; on the other, Leuridan ([2012]) argues that it should be understood as a special form of causal dependence. Craver and Bechtel ([2007]) offer two reasons to support its non-causal nature: (i) constitutive relevance holds among wholes and their parts, whereas causal relevance relates mereologically independent entities; and (ii) changes in the macro- and micro-behaviours, of a mechanism temporally overlap, whereas causes temporally precede their effects. The vast majority of mechanists join Craver and Bechtel in taking (i) and (ii) to establish the non-causal nature of constitutive relevance. However, Leuridan ([2012]) succeeds in making a powerful case to the contrary. He argues that if causation is understood along the lines of Woodward's interventionist theory of causation, which is the theory many mechanists adopt (Craver [2007b], in particular), it can be inferred that constitutive relevance is merely a form of bi-directional causal dependence.

Since the question of whether constitutive relevance is assumed to differ from causal relevance will be crucial to our argument, we provide that assumption with a label as well:

NC: Constitutive relevance is a non-causal form of dependence.

As NC corresponds to the majority opinion, we will standardly assume NC. We shall, however, explicitly consider the ramifications of rejecting that assumption.

Overall, then, we take the relation of constitutive relevance to be characterized by SUP, SC, NR, and NC. Applied to our exemplary mechanism α in Figure 1 that amounts to the following:

- (i) Ψ supervenes on $V_1 = \{\Phi_1, \Phi_2, \Phi_3\}$;
- (ii) the variables in V_1 model the behaviours of entities that are spatiotemporal parts of the entity and behaviours modelled by Ψ ;
- (iii) the variables in V_1 model non-redundant difference-makers of Ψ ;
- (iv) Ψ and the variables in V_1 are not causally related.

3 Mutual Manipulability and Interventionism

In light of the previous section, the problem of identifying the constituents of a mechanism can be more specifically characterized as the problem of detecting its spatio-temporal parts that are non-redundant difference-makers of the mechanism's macro-behaviour such that the latter supervenes on the

former. Currently, the most widely adopted approach for identifying relations of constitutive relevance is Craver's ([2007a], [2007b]) mutual manipulability approach MM:

My claim is that to establish that X 's Φ -ing is relevant to S 's Ψ -ing it is sufficient that one be able to manipulate S 's Ψ -ing by intervening to change X 's Φ -ing (by stimulating or inhibiting) and that one be able to manipulate X 's Φ -ing by manipulating S 's Ψ -ing. To establish that a component is irrelevant, it is sufficient to show that one cannot manipulate S 's Ψ -ing by intervening to change X 's Φ -ing and that one cannot manipulate X 's Φ -ing by manipulating S 's Ψ -ing. (Craver [2007a], p. 17; see also Craver [2007b], p. 159)

MM stipulates that if behaviour Ψ of S (modelled by $\Psi(S)$) and behaviour Φ_i of its spatio-temporal part X_i ($\Phi_i(X_i)$) are mutually manipulable, then $\Phi_i(X_i)$ is constitutively relevant to $\Psi(S)$; and if $\Psi(S)$ and $\Phi_i(X_i)$ are not mutually manipulable, then $\Phi_i(X_i)$ is not constitutively relevant to $\Psi(S)$. Taken in combination, MM provides a sufficient and necessary condition for constitutive relevance.⁴ Furthermore, Craver ([2007b], Section 4.8.3) understands manipulability in terms of the existence of a possible ideal intervention as defined by Woodward ([2003], p. 98).⁵ In sum, MM amounts to this:

MM: $\Phi_i(X_i)$ is constitutively relevant to $\Psi(S)$ if and only if (i) X_i and its behaviour Φ_i are spatio-temporal parts of S and its behaviour Ψ ; (ii) there exists a possible intervention $\mathcal{I}_{\Phi_i} = i_{\Phi_i}$ on $\Phi_i(X_i)$ with respect to $\Psi(S)$ that changes $\Psi(S)$'s value (or probability distribution); and (iii) there exists a possible intervention $\mathcal{I}_{\Psi} = i_{\Psi}$ on $\Psi(S)$ with respect to $\Phi_i(X_i)$ that changes $\Phi_i(X_i)$'s value (or probability distribution).

The basic idea behind MM is that the mutual manipulability of S 's Ψ -ing and the spatio-temporally overlapping Φ_i -ing of X_i guarantees that X_i 's Φ_i -ing is a non-redundant difference-maker of S 's Ψ -ing. Moreover, the fact that this difference-making relation holds among spatio-temporally overlapping entities indicates that it is of a non-causal nature. Overall, the dependencies that are selected by MM are exactly those dependencies that have the features of constitutive relevance pinpointed in the previous section.

MM entails that in order to show that Φ_i is a constituent of Ψ , it must be shown that there exists a possible intervention $\mathcal{I}_{\Phi_i} = i_{\Phi_i}$ on Φ_i with respect to Ψ that is associated with changes in Ψ , and a possible intervention $\mathcal{I}_{\Psi} = i_{\Psi}$ on Ψ

⁴ Couch ([2011], p. 382) claims that Craver's MM only provides a sufficient condition for constitutive relevance. In our view, textual evidence clearly contradicts that assessment.

⁵ Craver ([2007b], p. 154) adapts Woodward's definition of an ideal intervention for the context of his presentation. However, as Leuridan ([2012], Appendix) has shown that all Craver-interventions are Woodward-interventions, this adaptation does not make a difference to our discussion. We will hence focus on the wording of Woodward's much more well-known definition. For a detailed comparison of Craver's and Woodward's definitions see (Leuridan [2012], Section 5).

with respect to Φ_i that is associated with changes in Φ_i . For instance, the release of potassium ions in a mimosa's pulvini cells is shown to be constitutively relevant to the mimosa's nastic movement if it is demonstrated that it is possible to intervene both on the pulvini cells, thereby affecting the mimosa's movement, and on the mimosa as whole, thereby affecting its pulvini cells.

The most straightforward way of establishing such a possibility is to come up with actual bottom-up and top-down interventions, $\mathcal{I}_{\Phi_i} = i_{\Phi_i}$ and $\mathcal{I}_{\Psi} = i_{\Psi}$, as required by MM, for all actual interventions are possible interventions. In the case of the mimosa, the introduction of potassium-channel blockers is a candidate bottom-up intervention, whereas touching the mimosa by hand is a candidate top-down intervention. By contrast, subject to MM, establishing that Φ_i is not a constituent of Ψ is more intricate. To this end, it must be shown that interventions of types $\mathcal{I}_{\Phi_i} = i_{\Phi_i}$ and $\mathcal{I}_{\Psi} = i_{\Psi}$ do not possibly exist. Such a negative existential can only be substantiated inductively or via some *a priori* (conceptual or theoretical) argument that rules out required interventions.

As indicated before, Woodward's ([2003]) interventionist account of causation, interventionism for short, constitutes the conceptual background of MM. Interventionism is currently one of the most popular theories of causation. Nonetheless, before we can apply MM to our exemplary mechanism α , we have to review the main tenets of interventionism because definitional details will be crucial for our discussion. We suspect that one of the main reasons why so many authors believe that MM can be fruitfully put to use when it comes to accounting for constitution is that they rely on a merely intuitive understanding of interventionism without taking the definitional foundation of that theory at face value. For that reason, we will review the relevant definitions and extract their pertinent consequences—even though these definitions are frequently cited in the literature.

The definitions that constitute the core of Woodward's interventionism are the following. (We slightly adapt Woodward's symbols to avoid confusions with the symbols used in this article.)

M: A necessary and sufficient condition for $\mathcal{X}$ to be a (type-level) direct cause of $\mathcal{Y}$ with respect to a variable set $\mathbf{V}$ is that there be a possible intervention on $\mathcal{X}$ that will change $\mathcal{Y}$ or the probability distribution of $\mathcal{Y}$ when one holds fixed at some value all other variables $\mathcal{Z}_i$ in $\mathbf{V}$. A necessary and sufficient condition for $\mathcal{X}$ to be a (type-level) *contributing cause* of $\mathcal{Y}$ with respect to variable set $\mathbf{V}$ is that (i) there be a directed path from $\mathcal{X}$ to $\mathcal{Y}$ such that each link in this path is a direct causal relationship [...] and that (ii) there be some intervention on $\mathcal{X}$ that will change $\mathcal{Y}$ when all other variables in $\mathbf{V}$ that are not on this path are fixed at some value. (Woodward [2003], p. 59)

IV: $\mathcal{I}$ is an intervention variable for $\mathcal{X}$ with respect to $\mathcal{Y}$ if and only if [...]
(i) $\mathcal{I}$ causes $\mathcal{X}$; (ii) $\mathcal{I}$ acts as a switch for all the other variables that cause $\mathcal{X}$ [...]; (iii) any directed path from $\mathcal{I}$ to $\mathcal{Y}$ goes through $\mathcal{X}$ [...]; (iv) $\mathcal{I}$ is

(statistically) independent of any variable Z that causes $\mathcal{Y}$ and that is on a directed path that does not go through $\mathcal{X}$. (Woodward [2003], p. 98)

Relative to the notion of an intervention variable, an intervention on $\mathcal{X}$ with respect to $\mathcal{Y}$ then amounts to an intervention variable $\mathcal{I}$ for $\mathcal{X}$ with respect to $\mathcal{Y}$ taking some value i such that $\mathcal{I} = i$ causes $\mathcal{X}$ to take some value x (Woodward [2003], p. 98).

For our purposes, two things need to be made explicit about a theory built on M and IV. First, the notion of causation provided by M is relativized to a set of variables $\mathbf{V}$, but the notion of an intervention variable defined by IV is not relativized in that manner (Woodward [2008], p. 202).⁶ $\mathcal{X}$ is a direct or contributing cause of $\mathcal{Y}$ only relative to some variable set $\mathbf{V}$, while the interventionist nature of a variable $\mathcal{I}$ does not depend on the set of modelled variables. It must also be noted that the notion of causation that appears in IV is not the relativized notion defined in M, that is, it is not the ternary relation ' $\mathcal{X}$ causes $\mathcal{Y}$ with respect to $\mathbf{V}$ '. Rather, IV draws on causation *simpliciter*, which Woodward ([2008], p. 209) defines via existential generalization of M: a variable $\mathcal{X}$ is a cause of $\mathcal{Y}$ if and only if there exists a set $\mathbf{V}$ with respect to which $\mathcal{X}$ is a direct or a contributing cause of $\mathcal{Y}$ as defined by M.

Second, M and IV establish a tight connection between manipulability, difference-making in context, and causality, which Woodward ([2003], p. 61) sums up in the following slogan: no causal difference without a difference in manipulability relations, and no difference in manipulability relations without a causal difference. In particular, the analysans of causation supplied by M stipulates that if $\mathcal{X}$ is a cause of $\mathcal{Y}$, then (i) there exists a possible intervention $\mathcal{I} = i$ on $\mathcal{X}$ with respect to $\mathcal{Y}$, and (ii) all other variables in the relevant variable set $\mathbf{V}$ that are not on a path from $\mathcal{X}$ to $\mathcal{Y}$ can be fixed when $\mathcal{X}$ is wiggled through $\mathcal{I} = i$. That is, M determines that the manipulability of $\mathcal{X}$ and the fixability of all off-path variables in $\mathbf{V}$ are each necessary for $\mathcal{X}$ to cause $\mathcal{Y}$.⁷

As we shall see below, Woodward ([forthcoming]) has meanwhile modified his original definitions. In order not to confuse causation and interventions as defined by M and IV with their counterparts as defined in (Woodward [forthcoming]), we shall subsequently speak of M-causation and IV-interventions. Moreover, we shall assume that our exemplary mechanism α is completely stable and that interventions on α produce invariant results. That means compensatory responses and indirect interferences are assumed to be cancelled by additional interventions and/or the experimental setup—which is a standard assumption the mutual manipulability approach requires (Craver [2007b], pp. 156–7).

⁶ If IV were relativized like M, interventionism could not distinguish between difference-making relations that stem from causal dependencies and difference-making relations that are due to common causes (for details, see Baumgartner [2013], pp. 12–3).

⁷ For further details on these implications of M and IV see (Baumgartner [2012]).

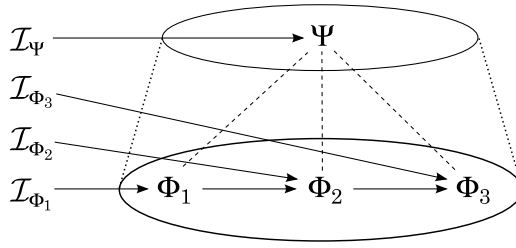


Figure 2. Test design induced by MM to identify the constitutive relations between Φ_1 , Φ_2 , Φ_3 , and Ψ .

We are now in a position to apply MM, M, and IV to α . MM determines that the elements of $\mathbf{V}_1 = \{\Phi_1, \Phi_2, \Phi_3\}$ are constituents of Ψ only if for every $\Phi_i \in \mathbf{V}_1$ there exists a possible intervention $\mathcal{I}_{\Phi_i} = i_{\Phi_i}$ on Φ_i with respect to Ψ , such that $\mathcal{I}_{\Phi_i} = i_{\Phi_i}$ is associated with changes in Ψ , and for every $\Phi_i \in \mathbf{V}_1$ there exists a possible intervention $\mathcal{I}_{\Psi} = i_{\Psi}$ on Ψ with respect to Φ_i , such that $\mathcal{I}_{\Psi} = i_{\Psi}$ is associated with changes in Φ_i . That is, in order for the elements of $\mathbf{V}_1$ to be constituents of Ψ , intervention variables as depicted in Figure 2 are required. However, in what follows we prove that such intervention variables are ruled out, and thus that the elements of $\mathbf{V}_1$ are entailed not to be constituents of Ψ by MM.

More specifically, we prove that top-down interventions for the macro-level of α that are associated with changes in α 's micro-level cannot possibly exist. To this end, we first identify a condition ($\dagger$), which, according to MM, is necessary for the elements of $\mathbf{V}_1 = \{\Phi_1, \Phi_2, \Phi_3\}$ to be constitutively relevant to Ψ and which requires the availability of suitable top-down intervention variables. Second we show that the latter are ruled out and, thus, that condition $\dagger$ is unsatisfiable. Here is condition $\dagger$:

$\dagger$: For every $\Phi_i \in \mathbf{V}_1$, there is an intervention variable $\mathcal{I}_{\Psi}$ for Ψ with respect to Φ_i , such that $\mathcal{I}_{\Psi} = i_{\Psi}$ (possibly) induces a change in both Ψ and Φ_i .

Now, assume (for *reductio*) that the elements of $\mathbf{V}_1$ are constituents of Ψ . As $\dagger$ is necessary for constitution, it follows that for every $\Phi_i \in \mathbf{V}_1$ there is an intervention variable $\mathcal{I}_{\Psi}$ for Ψ with respect to Φ_i . From this it follows by IV(i) that $\mathcal{I}_{\Psi}$ is a cause (*simpliciter*) of Ψ , which, in turn, means that there is a variable set $\mathbf{V}_2$ containing $\mathcal{I}_{\Psi}$ and Ψ , relative to which $\mathcal{I}_{\Psi}$ is an M-cause of Ψ . According to M, this entails that there is an intervention variable $\mathcal{I}_{\mathcal{I}_{\Psi}}$ for $\mathcal{I}_{\Psi}$ with respect to Ψ such that changes induced on $\mathcal{I}_{\Psi}$ via $\mathcal{I}_{\mathcal{I}_{\Psi}}$ are associated with changes in Ψ . SUP yields that all changes in Ψ are necessarily associated with changes in at least one $\Phi_j \in \mathbf{V}_1$, from which it follows that there is a variable set $\mathbf{V}_3$ containing $\mathcal{I}_{\Psi}$ and Φ_j with respect to which changes induced on $\mathcal{I}_{\Psi}$ via

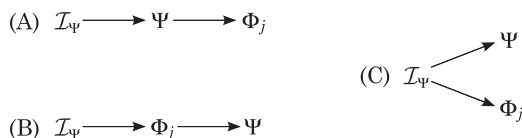


Figure 3. The three possible causal structures in which $\mathcal{I}_\Psi$ is a cause of both Ψ and an element Φ_j of $\mathbf{V}_1$.

$\mathcal{I}_{\mathcal{I}_\Psi}$ are associated with changes in Φ_j . Based on this, M yields that $\mathcal{I}_\Psi$ is an M-cause of Φ_j with respect to $\mathbf{V}_3$ and *ipso facto* a cause *simpliciter* of Φ_j . That is, $\mathcal{I}_\Psi$ is an M-cause of both Ψ and Φ_j . Figure 3 depicts the three possible causal structures in which $\mathcal{I}_\Psi$ is a cause of both Ψ and Φ_j .

In (A) and (B) of Figure 3, there is a causal relationship between Ψ and Φ_j (which is contained in $\mathbf{V}_1$). Thus, if NC is assumed, according to which the elements in $\mathbf{V}_1$ and Ψ are not causally related, it follows that the only structure that can accommodate the fact that $\mathcal{I}_\Psi$ is a cause of both Ψ and Φ_j is (C). That is, NC entails that $\mathcal{I}_\Psi$ causes Ψ and Φ_j along two different paths, or to put it another way, $\mathcal{I}_\Psi$ is a common cause of Ψ and Φ_j .

Now, manipulating Ψ via $\mathcal{I}_\Psi$, which, as shown above, shall be an intervention variable for Ψ with respect to Φ_i , either is associated with changes in Φ_i or it is not. If it is, then Φ_i is one of those $\Phi_j \in \mathbf{V}_1$ for which we demonstrated that they are caused by $\mathcal{I}_\Psi$ along a path that differs from the path along which $\mathcal{I}_\Psi$ causes Ψ (see structure (C) in Figure 3). Yet, subject to IV(iii), intervention variables for Ψ with respect to Φ_i must not be common causes of Ψ and Φ_i . That is, satisfying IV(i) in the vein of (C)—that is, the only NC-compatible way of satisfying IV(i)—yields that IV(iii) is violated, which stipulates that all directed causal paths from $\mathcal{I}_\Psi$ to Φ_i must go through Ψ . Thus, $\mathcal{I}_\Psi$ does not comply with all conditions of IV and, accordingly, cannot be an intervention variable for Ψ with respect to Φ_i after all. By contrast, if manipulating Ψ via $\mathcal{I}_\Psi$ is not associated with changes in Φ_i , then $\mathcal{I}_\Psi$ may count as an intervention variable for Ψ with respect to Φ_i , but one that is unserviceable in accounting for a constitutive relation between Ψ and Φ_i . In neither case does $\mathcal{I}_\Psi$ satisfy $\dagger$ for Ψ with respect to Φ_i .

Nothing in the above reasoning hinges on $\mathcal{I}_\Psi$ being our candidate intervention variable. The same reasoning can be repeated for any other candidate intervention variable. In the structure of mechanism α , it is either the case that IV(i) and IV(iii) cannot be satisfied together for a triple $\langle \mathcal{I}_\Psi, \Psi, \Phi_i \rangle$, where $\Phi_i \in \mathbf{V}_1$; or, if IV(i) and IV(iii) can be jointly satisfied for a triple $\langle \mathcal{I}_\Psi, \Psi, \Phi_i \rangle$, then Φ_i does not change when Ψ is wiggled via $\mathcal{I}_\Psi$. Therefore, $\dagger$ —which is necessary for constitution—cannot be satisfied for α . Thus, MM entails that the elements of $\mathbf{V}_1$ are not constituents of Ψ —which concludes our *reductio* of the initial assumption that the elements of $\mathbf{V}_1$ are constituents of Ψ .

If we take mechanism α to be instantiated in a mimosa's nastic movement, this *reductio* shows that touching the leaves of the mimosa, for example, does not count as an intervention on the mimosa with respect to its pulvini cells, because the touch does not surgically affect the mimosa on the macro-level. Rather, it is a common cause of the mimosa's movement and the potassium release in the pulvini cells. The same holds for any other cause of the mimosa's nastic movement. The latter can only be causally affected by common causes of the mimosa's macro- and micro-behaviours. None of these common causes are interventions in the sense of IV. Therefore, there cannot possibly exist any top-down interventions on a mimosa's nastic movement, which, subject to MM, is thus not constituted by the potassium release in the pulvini cells—nor by any other behaviour of its spatio-temporal parts.

Likewise, nothing in the above reasoning hinges on α being our exemplary mechanism. $\dagger$ cannot be satisfied for any mechanism. All that is required to establish that there cannot possibly exist an intervention variable as defined by M and IV for any macro-variable Ψ with respect to any micro-variable Φ_i , such that Φ_i changes when Ψ is wiggled, is that Φ_i is contained in a variable set V , such that Ψ supervenes on V and that the elements of V and Ψ are not causally related. That is, as long as constitutive relevance is taken to be characterized by SUP and NC, there cannot possibly exist M-IV-defined intervention variables for macro-variables with respect to their micro supervenience bases such that the latter could be changed by intervening on the former.⁸ But from the latter finding, MM infers that no spatio-temporal parts on which a mechanism's macro-level supervenes can be constitutively relevant to the latter. In sum, against the background of SUP, NC, M, and IV, MM implies INEX, namely, the inexistence of constitutive relations:

INEX: There do not exist any relationships of constitutive relevance between macro-phenomena and their spatio-temporal parts.

Thus, if MM is spelled out in terms of Woodward's original interventionism, for which Craver ([2007b]) explicitly designed it, it reduces the mechanistic paradigm to absurdity.

This *reductio* argument rests on the assumptions SUP, NC, M, and IV. To avoid its consequence, at least one of these assumptions must be discarded.

⁸ Based on a problematization of the satisfiability of IV(iii) that bears some similarity to ours, Glauer ([2012], p. 75) claims, first, that top-down interventions on Ψ with respect to Φ_i are not well-defined and, second, that bottom-up interventions on Φ_i with respect to Ψ are impossible due to a violation of IV(iii). We cannot follow Glauer on either count. Relative to a model of a mechanism as in Figure 1, it is precisely defined what a top-down intervention amounts to and bottom-up interventions on a mechanism may well exist, for SUP only entails that a mechanism's macro-level depends on its micro-level, but not the other way around. That means it is possible to intervene on the micro-level by an IV-intervention that is not associated with changes in the macro-level. What SUP rules out is the existence of IV-interventions on the macro with respect to the micro-level.

While SUP is uncontested, Leuridan ([2012]) rejects NC and Woodward ([forthcoming]) proposes weakened variants of M and IV. We discuss the consequences of rejecting NC in the remainder of this section and Woodward's newest variant of interventionism in the next section.

Leuridan's reason for rejecting the non-causal nature of constitution NC is that it is a direct consequence of Woodward's interventionist theory that IV-manipulability entails causation and, thus, mutual manipulability entails bi-directional causation. He argues that Craver cannot consistently embed MM within a Woodwardian framework and assume the non-causal nature of constitution as expressed in NC. Leuridan ([2012], p. 424) concludes: 'Causal relevance (*sensu latu*) thus comes in two varieties: intralevel etiological relevance (or causal relevance *sensu strictu*) and interlevel relevance'.

We agree with Leuridan that, if constitutive relevance is not assumed to be of non-causal nature, then MM, interpreted against the background of M and IV, implies that constitutive relevance reduces to a form of causal relevance. Moreover, discarding NC yields that the *reductio* argument leading to INEX is blocked, such that cases of constitution—or rather constitutive causation—are no longer entailed to be inexistent on mere conceptual grounds. To see this, note first that if we discard NC it still holds in mechanism α that all causes of Ψ are also causes of some $\Phi_i \in V_1$, for due to SUP all changes induced on Ψ are associated with changes in some $\Phi_i \in V_1$. But by abandoning NC, it no longer follows that Ψ and $\Phi_i \in V_1$ must be located on different causal paths. Rather, it becomes possible that Ψ and Φ_i are located on one and the same causal path. If Ψ and some $\Phi_i \in V_1$ are on the same path, IV(i) and IV(iii) are rendered satisfiable at the same time. It thereby becomes possible to intervene on Ψ with respect to Φ_i in the sense defined by IV, such that manipulations of Ψ are associated with changes in Φ_i . Hence, condition $\dagger$ turns out to be satisfiable.

If (and only if) for every element Φ_i of V_1 there exist possible interventions of type (A) on Ψ with respect to Φ_i (see Figure 3), such that Φ_i changes when Ψ is wiggled, and there exist possible interventions of type (B) on Φ_i with respect to Ψ , such that Ψ changes when Φ_i is wiggled, then MM yields that the elements of V_1 and Ψ are related in terms of constitution (or constitutive causation). Relationships of constitutive causation are always bi-directional, that is, they amount to causal loops. Overall, if Ψ and the elements of V_1 are mutually manipulable, the causal structure entailed by SUP, M, IV, and MM for α is the one depicted in Figure 4.

The resulting causal structure features numerous causal cycles. Note that the causal dependencies in Figure 4 hold among wholes and their parts, that changes in Ψ and changes in V_1 temporally overlap, and that the causal influence is transmitted instantaneously along the interlevel paths. All of these features of the structure in Figure 4 are very unusual for causal structures. If NC is discarded, the resulting structures of constitutive

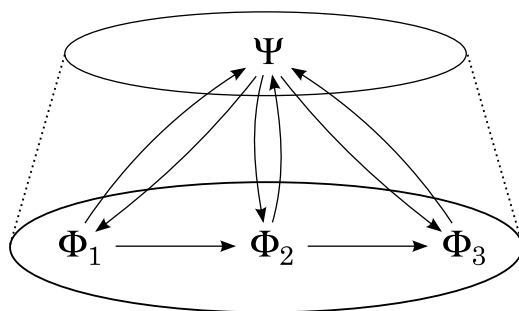


Figure 4. Causal structure underlying mechanism α , provided that constitution is not assumed to be of a non-causal nature.

causation differ from ordinary causal structures in numerous crucial respects. In light of this, the vast majority of mechanists do not join Leuridan ([2012]) in abandoning NC. Fortunately, there is another way to block the *reductio* argument leading to INEX.

4 Modifying Interventionism

The problem of MM revealed in the previous section is but one problem among many that Woodward's ([2003]) theory gives rise to when applied to variables that are dependent in non-causal ways (for another such problem, see Baumgartner [2010], [2013]). Woodward ([forthcoming]) has recently offered a modified version of interventionism, which he claims avoids the problems associated with non-causal dependence relations. In this section, we reconsider MM when read against the background of this most recent proposal.

Woodward's ([forthcoming]) basic idea for modifying his original definitions is to introduce exemption clauses for non-causal dependence relations. He proposes the amended definitions M^* and IV^* , given below. In both of them, the notion of supervenience appears as proxy for all other non-causal dependence relations. For brevity, we merge Woodward's twofold account of direct and contributing causation into one notion of causation M^* :

M^* : $\mathcal{X}$ is a cause of $\mathcal{Y}$ with respect to variable set $\mathbf{V}$ if and only if there possibly exists an IV^* -intervention on $\mathcal{X}$ with respect to $\mathcal{Y}$, when all variables in $\mathbf{V}$ that are not on a causal path from $\mathcal{X}$ to $\mathcal{Y}$ and are not related in terms of supervenience to $\mathcal{X}$ or $\mathcal{Y}$ are held fixed, such that the value or the probability distribution of $\mathcal{Y}$ changes.⁹

⁹ Woodward ([forthcoming]) does not state M^* explicitly, but merely indicates its relevant features. M^* is our reconstruction of Woodward's suggestion. For a justification as to why this is an adequate reconstruction, see Baumgartner ([2014], Section 3).

IV*: $\mathcal{I}$ is an intervention variable for $\mathcal{X}$ with respect to $\mathcal{Y}$ if and only if $\mathcal{I}$ satisfies IV(i), IV(ii), IV(iii*), and IV(iv*)

IV(iii*): any directed path from $\mathcal{I}$ to $\mathcal{Y}$ goes through $\mathcal{X}$ or through a variable $\mathcal{Z}$ which is related to $\mathcal{X}$ in terms of supervenience;

IV(iv*): $\mathcal{I}$ is (statistically) independent of every cause of $\mathcal{Y}$ that is neither located on a path through $\mathcal{X}$ nor on a path through a variable $\mathcal{Z}$ which is related to $\mathcal{X}$ in terms of supervenience.

To distinguish this modified variant of interventionism from the original variant, we shall speak of M^* -causation and IV^* -interventions and we shall say that M^* and IV^* constitute interventionism*.

The relevant respects (for our purposes) in which the exemption clauses in M^* and IV^* return definitions that differ from M and IV are as follows: First, according to IV^* , a variable $\mathcal{I}$ can count as an intervention variable for $\mathcal{X}$ with respect to $\mathcal{Y}$ even if $\mathcal{I}$ is connected to $\mathcal{Y}$ through a path that does not go through $\mathcal{X}$, that is, through what we shall call an $\overline{\mathcal{X}}$ -path, provided that this $\overline{\mathcal{X}}$ -path goes through a variable contained in the supervenience base of $\mathcal{X}$. Second, M^* entails that in order to establish that $\mathcal{X}$ is a cause of $\mathcal{Y}$, those variables that are contained in the supervenience bases of $\mathcal{X}$ and $\mathcal{Y}$ do not have to be fixed.

Let us now see what the implications of MM are when read against the background of interventionism*. Again, we first apply MM to mechanism α and assume all the features assigned to constitution in Section 2, and we once more assume that the variables in $V_1 = \{\Phi_1, \Phi_2, \Phi_3\}$ are constituents of Ψ . From this, it follows by MM —more specifically, by condition $\dagger$ entailed by MM —that for every $\Phi_i \in V_1$ there is an intervention variable $\mathcal{I}_\Psi$ for Ψ with respect to Φ_i . $\mathcal{I}_\Psi$ is a cause of Ψ , which means there is a variable set V_2 containing $\mathcal{I}_\Psi$ and Ψ , relative to which there is an intervention variable $\mathcal{I}_{\mathcal{I}_\Psi}$ for $\mathcal{I}_\Psi$ with respect to Ψ such that changes induced on $\mathcal{I}_\Psi$ via $\mathcal{I}_{\mathcal{I}_\Psi}$ are associated with changes in Ψ . Due to SUP , all changes in Ψ are necessarily associated with changes in at least one $\Phi_j \in V_1$, meaning there is a variable set V_3 relative to which $\mathcal{I}_\Psi$ is an M^* -cause and, thus, a cause (*simpliciter*) of Φ_j . That is, $\mathcal{I}_\Psi$ is a cause of both Ψ and Φ_j . Of all possible causal structures in Figure 3 that feature $\mathcal{I}_\Psi$ as cause of both Ψ and some elements of V_1 , only (C) is compatible with NC . Hence, $\mathcal{I}_\Psi$ must be a common cause of Ψ and Φ_j .

Up to this point the consequences of MM , when embedded in interventionism*, are identical to its consequences when embedded in interventionism. But while $\mathcal{I}_\Psi$ in structure (C) of Figure 3 cannot be an IV -intervention variable for Ψ with respect to those elements of V_1 that change due to SUP when intervening on Ψ , $\mathcal{I}_\Psi$ in (C) can be such an IV^* -intervention variable for Ψ . The reason is that IV^* allows for $\overline{\Psi}$ -paths from $\mathcal{I}_\Psi$ to elements of V_1 , provided that these $\overline{\Psi}$ -paths go through the supervenience base of Ψ —which is the case in structure (C). In

consequence, Ψ is manipulable via an IV*-intervention on Ψ with respect to some $\Phi_i \in V_1$, such that Φ_i changes its value. Moreover, if it turns out that for every $\Phi_i \in V_1$ we find an IV*-intervention of type (C) on Ψ with respect to Φ_i , such that Φ_i changes, Ψ is manipulable with respect to every element of V_1 . Finally, it is also possible that for every $\Phi_i \in V_1$ we find an IV*-intervention on Φ_i with respect to Ψ such that Ψ changes. In that case, MM is applicable and identifies the elements of V_1 as constituents of Ψ . So far so good.

However, note that if Ψ and the elements of V_1 turn out to be mutually manipulable, not only is MM applicable, but also M^* . If Ψ is IV*-manipulable with respect to every $\Phi_i \in V_1$ and every $\Phi_i \in V_1$ is IV*-manipulable with respect to Ψ , M^* rules that there is a causal feedback between Ψ and every $\Phi_i \in V_1$. The reason is that M^* does not require supervenience bases to be held fixed when macro-variables are manipulated. Yet, NC entails that Ψ is causally independent of all its constituents in V_1 . In sum, applying MM to α against the background of SUP, NC, M^* , and IV* results in a contradiction: Ψ and the elements of V_1 are causally dependent, which is entailed by M^* , and not causally dependent, which is entailed by MM and NC.

Again, nothing in the above reasoning hinges on our particular choice of variables or on the concrete structure of our exemplary mechanism. SUP, M^* , and IV* yield that macro-variables and their constituents are causally dependent, whereas MM and NC entail that they are independent. Embedding MM in interventionism* reduces the mechanistic framework to absurdity in a way that is closely related to the *reductio* of MM when embedded in interventionism. While, in the latter case, assuming the variables in V_1 to be constituents of Ψ entails that they are not constituents of Ψ , the same assumption, in the former case, entails that the elements of V_1 are causally relevant to Ψ and not causally relevant to Ψ . As in the latter case, not only the non-causal nature of constitution, but also the assumption of the existence of constitutive relationships, are candidates for rejection when MM is embedded in interventionism*. But discarding the existence of constitutive relationships would just yield INEX again. If we want to insist that mechanists do not chase a chimera and if we want constitution to be a non-causal form of dependence, modifying M and IV in terms of M^* and IV* does not solve the problem revealed in the previous section. Interventionist definitions must be modified further.

One may conceive of several supplementary modifications of interventionist definitions. In what follows, we propose our preferred option without claiming that this is the only way to go. As we want to uphold MM, we have to block the second *reductio* argument presented above by preventing the mutual manipulability of a mechanism's macro- and micro-level entailing (mutual) causation. A straightforward way to do so is by stipulating that spatio-temporally overlapping entities and their behaviours are not causally related, even if they are mutually manipulable. We propose to build a constraint into M^* to

the effect that, in order for $\mathcal{X}$ to be a cause of $\mathcal{Y}$, it must be possible to intervene on $\mathcal{X}$ to change the value of $\mathcal{X}$ at a time t such that $\mathcal{Y}$ changes at a (strictly) later time t' .¹⁰

M:** $\mathcal{X}$ is a cause of $\mathcal{Y}$ with respect to variable set $\mathbf{V}$ if and only if there exist times t, t' , where $t < t'$, such that there possibly exists an IV*-intervention on $\mathcal{X}$ with respect to $\mathcal{Y}$ that sets $\mathcal{X}$ to some value x at t , when all variables in $\mathbf{V}$ that are not on a causal path from $\mathcal{X}$ to $\mathcal{Y}$ and are not related in terms of supervenience to $\mathcal{X}$ or $\mathcal{Y}$ are held fixed, such that the value or the probability distribution of $\mathcal{Y}$ changes at t' .

t and t' must be understood to stand for more or less extended time intervals such that t is strictly before t' . Note that M** is only a sparse modification of M*: it does not stipulate that changes in causes always strictly precede changes in their effects. Rather, it merely requires that at least one possible intervention on a cause induces a change in that cause which strictly precedes a change in the corresponding effect. Nonetheless, such a sparse modification suffices to guarantee that relationships of mechanistic constitution are entailed to be of non-causal nature.

To see this, reconsider our exemplary mechanism α . As before, variables that are common causes of Ψ and of some $\Phi_i \in \mathbf{V}_1$ count as IV*-intervention variables for Ψ with respect to Φ_i . That is, even if for all $\Phi_i \in \mathbf{V}_1$ it holds that every intervention variable $\mathcal{I}_\Psi$ on Ψ with respect to Φ_i is a common cause of Ψ and Φ_i , $\dagger$ is satisfiable for α . Accordingly, Ψ and the elements of $\mathbf{V}_1$ are rendered mutually manipulable, which, according to MM, induces constitution. At the same time, however, that particular form of mutual manipulability does not satisfy the right-hand side of M**, for, according to SC, all changes induced on Ψ temporally overlap with the changes in Φ_1 to Φ_3 . (For instance, the drop in turgor pressure of a mimosa's pulvini cells always coincides with the nastic movement of the plant.) Or the other way around: for no $\Phi_i \in \mathbf{V}_1$ there exist two strictly sequential times t and t' such that Ψ is changed at t and Φ_i changes at t' , or Φ_i is changed at t and Ψ changes at t' . M** thus entails that Ψ is causally independent of all variables in $\mathbf{V}_1$. These findings generalize for all mechanisms and all chosen variable sets: as SC is assumed to hold for all constitutive relations, the right-hand side of M** is unsatisfiable, in principle, by macro- and micro-levels of mechanisms.

Overall, if we embed MM in interventionism**, namely, the union of M** and IV*, we—for the first time—get a consistent account of mechanistic constitution, which does not force INEX upon us. What is more, the resulting

¹⁰ We do not want to claim that M** amounts to an account of causation that produces adequate results in every conceivable context of application. Our claim is merely that if one wants to embed MM in a variant of interventionism that accounts for causation as it figures in mechanisms, M** is a promising way to go.

account entails the non-causal nature of constitution NC. Cashing out MM within interventionism** dispenses us from additionally assuming NC. Both of these features are clear advantages the aggregate of MM and interventionism** has over any other currently available interventionist embedding of Craver's mutual manipulability criterion.

5 Fat-Handedness

Despite the progress we have made so far, there remains a severe problem. It is one of the main selling points of MM that it allegedly reproduces experimental practices in the sciences (Craver [2007b], Section 4). MM is taken to theoretically ground a method for empirically detecting relationships of constitutive relevance. Applied to our exemplary mechanism α , this amounts to the claim that the observed mutual manipulability of Ψ and the variables in V_1 provides empirical evidence in favour of the variables in V_1 being the constituents of Ψ .

Yet, while the embedding of MM in interventionism**—unlike the original embedding in interventionism—generates a consistent theory, it fails to account for mechanistic constitution on evidence-based grounds. To see this, reconsider mechanism α . In order to yield that the elements of V_1 are constituents of Ψ , MM requires that for every $\Phi_i \in V_1$ there exists an intervention on Ψ with respect to Φ_i , as well as an intervention on Φ_i with respect to Ψ , such that these interventions are associated with changes in both Ψ and Φ_i . In virtue of M**, all of these interventions are causes of both Ψ and Φ_i . As the relation between Ψ and the elements of V_1 is determined to be of non-causal nature by M**, all of these interventions required to satisfy mutual manipulability are entailed to be common causes of Ψ and Φ_i . MM can only be satisfied by fat-handed interventions on macro- and micro-levels,¹¹ which only count as interventions because IV* has built-in exemption clauses for supervenience relationships. In light of this result, however, the correlations of Ψ and the elements of V_1 that result from mutual manipulability do not need to be due to constitutive dependencies. These correlations could simply be due to the fact that all MM-induced interventions on α are fat-handed.

More concretely, Figures 5a and b depict the type of IV*-interventions that give rise to the mutual manipulability of the macro- and micro-levels of mechanism α . If we find correlations of Ψ and the micro-variables in V_1 under such fat-handed manipulations, there is no need to postulate constitutive dependencies. Such correlations can be accounted for by the mere fact that the macro- and micro-levels of α are systematically wigged with a fat hand.

¹¹ A fat-handed intervention is an intervention that influences its ultimate effect along two (or more) different causal paths (Scheines [2005], p. 932).

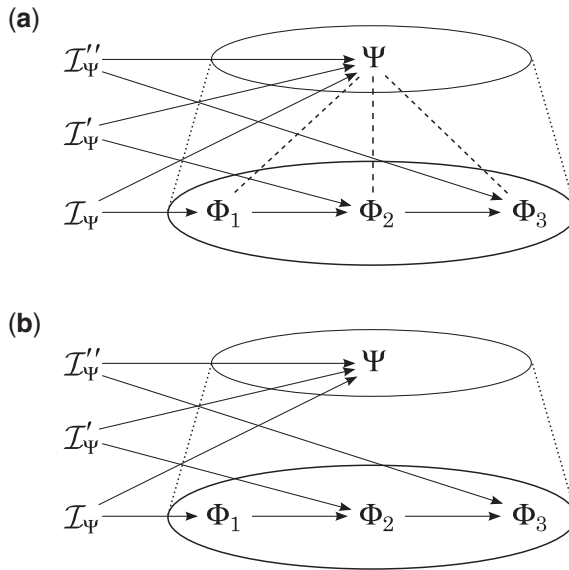


Figure 5. Two empirically indistinguishable models of mutual IV*-manipulations of α 's macro- and micro-levels. While the model (a) features relations of constitutive relevance, model (b) does not feature such relations.

The model in Figure 5(a), which features constitutive dependencies among Ψ and the elements of V_1 , and the model in Figure 5(b), which does not feature such dependencies, imply the same correlations under manipulations. They are empirically indistinguishable.

This result again generalizes for all mechanisms. M^{**} entails that macro- and micro-levels are only mutually manipulable via common causes. But then, the fact that they behave in a highly correlated manner can be explained by the mere fat-handed nature of corresponding mutual manipulations. Mutual manipulability via common cause interventions provides no empirical evidence in favour of the existence of constitutive dependencies. Thus, MM is not sufficient to account for constitution on evidence-based grounds.

In the remainder of this article, we make a suggestion how MM could be supplemented in a way that produces at least indirect empirical evidence—or second-order evidence—in favour of constitutive dependencies. To generate evidence for the existence of an additional dependence among macro- and micro-levels of a mechanism, top-down interventions are of core relevance. There exists an important asymmetry between bottom-up and top-down interventions: while it is possible to intervene on the macro-level of a mechanism without inducing changes on the macro-level, SUP renders it impossible to change the macro-level of a mechanism without thereby inducing changes on the micro-levels. Subject to SUP and M^{**} , every cause of a mechanism's

macro-level is a common cause of the macro-level and at least one element of the corresponding supervenience base.

In case of mechanism α , this means that every cause of Ψ is a common cause of Ψ and at least one $\Phi_i \in \mathbf{V}_1$. Now, suppose that we find a first cause $\mathcal{I}_\Psi$ of Ψ and suppose that $\mathcal{I}_\Psi$ turns out to be a (direct) common cause of Ψ and Φ_1 . (See, again, Figure 5 for an illustration.) Next, suppose that we expand the analysed variable set and that we find a second cause $\mathcal{I}'_\Psi$ of Ψ , which happens to be a (direct) common cause of Ψ and Φ_2 . Furthermore, a third cause $\mathcal{I}''_\Psi$ of Ψ turns out to be a (direct) common cause of Ψ and Φ_3 , a fourth cause $\mathcal{I}'''_\Psi$ turns out to be a (direct) common cause of Ψ and Φ_1 , and so on; the effect is that all causes of Ψ are common causes of type $\mathcal{I}_\Psi$, $\mathcal{I}'_\Psi$, or $\mathcal{I}''_\Psi$ in Figure 5. The resulting correlations can either be modelled by means of a structure featuring constitutive dependencies, as depicted in Figure 5a, or by means of a pure common cause model, as depicted in Figure 5b.

Even though the models in Figure 5 are equivalent with respect to implied correlations, the model in Figure 5(a) has a distinctive advantage over that in Figure 5(b): Figure 5(a) not only explains the correlations of Ψ and the corresponding variables in $\mathbf{V}_1$, it also explains why we do not find causes of Ψ that are not common causes of Ψ and some $\Phi_i \in \mathbf{V}_1$. The model in Figure 5(b) only accounts for the correlations of Ψ and the corresponding variables in $\mathbf{V}_1$, but it provides no account of why we do not at some point find a surgical cause of Ψ . If the underlying causal structure were a mere common-cause structure, expansions of analysed variable sets should be expected to yield causes of Ψ that are not common causes of Ψ and some $\Phi_i \in \mathbf{V}_1$. However, if the elements of $\mathbf{V}_1$ are constituents of Ψ , there cannot exist a surgical cause of Ψ . The constitutive model in Figure 5(a) has higher explanatory power than the common cause model in Figure 5(b). It not only explains resulting correlations, but it also explains why there do not exist surgical causes of Ψ .

Empirical evidence does not only consist in correlational evidence resulting from suitable manipulations. Expansions of analysed variable sets give rise to a sort of second-order evidence. If it turns out that all causes of Ψ we find are common causes of Ψ and some $\Phi_i \in \mathbf{V}_1$, we have good abductive reasons to prefer models that account for this second-order evidence by introducing additional dependencies among Ψ and the variables in $\mathbf{V}_1$. Models featuring constitutive relevance relations provide the best available explanation for an MM-satisfying system whose macro-behaviour can only be manipulated with a fat hand. That is, in order to establish that the elements of $\mathbf{V}_1$ are constitutively relevant to Ψ , we not only have to establish that Ψ and the variables in $\mathbf{V}_1$ are mutually manipulable, but we moreover have to establish that all causes of Ψ are common causes of Ψ and some $\Phi_i \in \mathbf{V}_1$.

Hence, the empirical evidence that justifies identifying the processes in a mimosa's pulvini cells as constituents of the plant's nastic movement does

not simply consist in singular mutual manipulations of the mimosa's macro- and micro-levels. The result of any single such experimental manipulation can be thoroughly accounted for in terms of a pure common cause model. Rather, what empirically justifies the inference to constitutive relevance is the fact that all further macro causes of the mimosa's nastic movement that we happen to find in additional experimental setups turn out to be common causes of the mimosa's macro-behaviour and some micro occurrence in its pulvini cells, that is, the fact that no cause of the nastic movement can be found that does not affect the pulvini cells in some way or another.

To put this in more explicit and in general terms, we introduce the criterion of fat-handedness FH:

FH: The elements of a set $V = \{\Phi_1, \Phi_2, \dots, \Phi_n\}$ and a variable Ψ satisfy FH if and only if every IV*-intervention on Ψ is a common cause of Ψ and some $\Phi_i \in V$.

We submit that the conjunction of MM and FH—embedded in interventionism**—amounts to a consistent account of constitutive relevance that accounts for constitutive relevance on evidence-based grounds:

CR: The elements of a set $V = \{\Phi_1, \Phi_2, \dots, \Phi_n\}$ are constitutively relevant to a variable Ψ if and only if the relationship between the elements of V and Ψ complies with MM and FH.

In sum, to empirically establish relations of constitutive relevance, establishing the mutual manipulability of a variable Ψ and the variables in a set V on the basis of a few IV*-interventions is not sufficient. Additionally, it must be shown that the dependencies between Ψ and V induced by interventions on Ψ cannot be screened off by suitable surgical interventions. Only after a systematic expansion of the analysed variable set that generates a sufficient second-order evidential basis to inductively infer that Ψ and the elements of V satisfy FH can it justifiably be concluded that Ψ and V 's elements are related in terms of constitutive relevance. It must be emphasized that the universal logical form of FH, which imposes constraints on every IV*-intervention on Ψ with respect to some Φ_i , yields that compliance with CR can, in principle, only be established inductively and is, thus, always prone to error.

6 Conclusion

The first part of this article has shown that Craver's ([2007a], [2007b]) mutual manipulability account of mechanistic constitution requires a theoretical embedding that is much more difficult to come by than is recognized by most mechanists. Embedding MM either in Woodward's ([2003]) original variant of interventionism—as suggested by Craver himself—or in Woodward's ([forthcoming]) recent interventionism*, while at the same time

assuming the non-causal nature of constitution NC, undermines the mechanistic framework by implying the inexistence of constitutive relationships INEX. One way out of the impasse would be to reject NC—that is, to accept, as does Leuridan ([2012]), that constitution is a special form of causal dependence. However, most friends of mechanistic constitution join Craver and Bechtel ([2007]) in contending that constitution differs in essential respects from causal dependence. That, in turn, means that the interventionist theory in which MM is embedded must be modified further.

In the second part, we proposed a temporal relativization of M^* —namely, M^{**} —which for the first time resulted in a consistent theoretical embedding of MM that neither forces INEX nor the rejection of NC upon us. Still, it turned out that this embedding in interventionism** falls short of accounting for constitutive relationships on evidence-based grounds, which, after all, is believed to be one of the core advantages of MM. Hence, we suggest using unresolvable fat-handedness of top-down interventions as an additional criterion, FH, which, in combination with MM, allows for generating (second-order) abductive evidence for constitutive dependencies.

Let us end with a caveat. We do not claim that the combination of MM, interventionism**, and FH accounts for how the notion of constitutive relevance is actually used in scientific practice. Establishing this would require successful replications of case studies based on our account, which is a project we have not even begun to undertake here. Rather, we contend that embedding MM in interventionism** and supplementing it with FH constitutes a consistent and empirically contentful account of mechanistic constitution, which allows for treating constitution and causation as two closely related, but still distinct, types of dependencies. According to this account, both constitution and causation are difference-making relations that stem from manipulability and control; but while the former is characterized by unresolvable fat-handedness and systematic temporal overlap, the latter allows for surgical interventions and, at least sometimes, features strict sequentiality.

The final upshot of this article is a call for philosophical caution. Craver's mutual manipulability account of constitution and Woodward's interventionist account of causation have a considerable amount of pre-theoretic intuitive plausibility. They both seem to directly capture scientific practice. But intuitive appeal and echoing scientific practice do not suffice for a fruitful, nor even for a consistent, philosophical theory.

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